

Constitutive Continuation Capacity

A machine-checked framework for prospective health

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Paper I in a series on the formal and measurement foundations of health.

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Abstract

We formalize prospective health as

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t)).}$$

Here $\mathfrak{S}$ supplies dynamics, constitutive semantics, a capacity representation, and a formal requirement language; r, d, t remain explicit indices.

The upstream response semantics are

$$\boxed{\mathcal{L}_{d,t}(x) \xrightarrow{\text{constitutive restriction}} \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\} \simeq \mathcal{V}_{d,t}(x).}$$

The completeness claim applies to the response family. Capacity is a downstream representation,

$$\boxed{\text{Capacity}_{\mathfrak{S}}(x; d, t) = \text{collect}(\{o(v) \mid v \in \mathcal{V}_{d,t}(x)\}),}$$

and may compress the viable-response landscape. In the constitutive specialization, a two-sided response representation is a conservative extension whose positive projection recovers this established capacity exactly. A finite obstruction separates positive persistence information from robust-persistence status.

A formal requirement language $\mathcal{Q}$ induces

$$c_1 \sim_{\mathcal{Q}} c_2 \iff \forall r \in \mathcal{Q}, \text{Adequate}_r(c_1) \leftrightarrow \text{Adequate}_r(c_2), \quad \boxed{\text{HealthCap}_{\mathcal{Q}} = \text{Cap}/\sim_{\mathcal{Q}} .}$$

Every adequacy predicate and associated health query descends to this quotient. Every requirement-sufficient encoding uniquely determines the quotient on its attained range. Finite witnesses show both strict collapse of requirement-invisible capacity variation and retention of requirement-visible differences.

A single state-changing fixture proves requirement, scenario, and horizon relativity. Qualified presentation invariance is established under explicit realization, capacity, and adequacy transport. Two boundary results delimit the foundation:

$$\boxed{\begin{array}{l} \text{positive prospective capacity} \not\Rightarrow \text{failure-sensitive robustness in general,} \\ \text{present-realization observation} \not\Rightarrow \text{sufficient observation of prospective health.} \end{array}}$$

Scientific licensing, richer risk/control semantics, and empirical identification form downstream layers that retain the foundational prospective-health schema.

1 Introduction: prospective health as organized future capacity

We study health as a property of a presently realized organization together with its capacity to continue adequately under specified future conditions.

Let x be a present state, d a scenario, t a horizon, r a formal health requirement, and $\mathfrak{S}$ a health specification. The foundational proposition is

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t)).} \quad (1)$$

Equation (1) resolves the judgment into two typed factors:

$$\boxed{\text{Realizes}_{\mathfrak{S}}(x) \quad (\text{present constitution}), \quad \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}) \quad (\text{future adequacy}).}$$

Their conjunction assigns prospective health to a presently instantiated organization with requirement-adequate continuation capacity.

The response-to-health chain is

$$\boxed{\begin{array}{l} \mathcal{L}_{d,t}(x) \xrightarrow{\text{constitutive restriction}} \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\} \simeq \mathcal{V}_{d,t}(x), \\ \mathcal{V}_{d,t}(x) \xrightarrow{\text{observe}} \text{Obs}_{d,t}(x) \xrightarrow{\text{collect}} \text{Capacity}_{\mathfrak{S}}(x; d, t) \xrightarrow{[-]_{\mathcal{Q}}} \text{HealthCap}_{\mathcal{Q}}. \end{array}} \quad (2)$$

The exact proof-carrying relation is the injective forgetful map:

$$\boxed{\begin{array}{l} \iota : \mathcal{V}_{d,t}(x) \hookrightarrow \mathcal{L}_{d,t}(x), \\ \text{im } \iota = \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\}, \quad \mathcal{V}_{d,t}(x) \simeq \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\}. \end{array}} \quad (3)$$

Thus the *complete* object is the viable-continuation family. The capacity is the specification's representation of that family after observation and collection.

The semantics of the foundational capacity are determined by the declared observer and collector:

$$\boxed{\text{Capacity}_{\mathfrak{S}}(x; d, t) = \text{collect}(\{o(v) \mid v \in \mathcal{V}_{d,t}(x)\}).} \quad (4)$$

A domain-specific observer and collector may enrich this positive representation with structure carried by viable histories, including costs or probabilities attached to successful continuations. Failure-sensitive quantities use a richer upstream response representation that retains the relevant lawful failures. Section 10 supplies the finite obstruction.

For a fixed formal requirement language $\mathcal{Q}$,

$$\boxed{\sigma_{\mathcal{Q}}(c)(r) = \text{Adequate}_r(c), \quad c_1 \sim_{\mathcal{Q}} c_2 \iff \sigma_{\mathcal{Q}}(c_1) = \sigma_{\mathcal{Q}}(c_2),} \quad (5)$$

and

$$\boxed{\text{HealthCap}_{\mathcal{Q}} = \text{Cap} / \sim_{\mathcal{Q}}.} \quad (6)$$

A single finite state-changing model supplies the principal controls:

$$\begin{array}{l} \neg \text{Health}_{\mathfrak{S},\text{stable}}(P; \text{mild}, \text{now}), \\ \text{Health}_{\mathfrak{S},\text{stable}}(P; \text{mild}, \text{long}), \\ \neg \text{Health}_{\mathfrak{S},\text{stable}}(P; \text{severe}, \text{long}), \\ \text{Health}_{\mathfrak{S},\text{any}}(P; \text{mild}, \text{now}). \end{array}$$

Requirement, scenario, and horizon are therefore mathematically active on the same present state.

Paper I establishes (1), its response semantics, its requirement-visible abstraction, and two immediate information boundaries. Later papers govern the admissible query family, enrich future-response structure, and study empirical identification while retaining the foundational normal form.

2 Health specifications and prospective health

Let X, D, T denote state, scenario, and horizon types. A health specification supplies

$$\mathfrak{S} = (\Phi, \mathcal{C}, O, \mathcal{Q}, \text{Adequate}), \quad \Phi = (\text{Hist}, \text{Lawful}),$$

with

$$\text{Hist} : (d, t, x, y) \mapsto \text{Type}, \quad \text{Lawful} : \text{Hist}(d, t, x, y) \rightarrow \text{Prop},$$

$$\mathcal{C} = (\text{Realizes} : X \rightarrow \text{Prop}, \text{Continues} : \text{Hist}(d, t, x, y) \rightarrow \text{Prop}),$$

and

$$O = (\text{Atom}, \text{Cap}, o, \text{collect}).$$

For fixed (d, t, x) ,

$$\mathcal{V}_{d,t}(x) := \{(y, \gamma) \mid \gamma \in \text{Hist}(d, t, x, y), \text{Lawful}(\gamma), \text{Continues}(\gamma)\}. \quad (7)$$

Define

$$\text{Obs}_{d,t}(x) = \{a \mid \exists v \in \mathcal{V}_{d,t}(x), o(v) = a\}, \quad (8)$$

$$\text{Capacity}_{\mathfrak{S}}(x; d, t) = \text{collect}(\text{Obs}_{d,t}(x)). \quad (9)$$

The family in (7) ranges over every proof-carrying viable continuation admitted by the declared dynamics and constitution. Equations (8)–(9) permit many-to-one observation and collection, so the capacity carrier may represent several viable continuations by the same downstream value.

Each

$$r \in \mathcal{Q}$$

supplies

$$\text{Adequate}_r : \text{Cap} \rightarrow \text{Prop}.$$

Definition 2.1 (Prospective health).

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) : \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}}(x; d, t))}. \quad (10)$$

¹

The following implications remain domain-specific propositions:

$$\text{Lawful}(\gamma) \Rightarrow \text{Continues}(\gamma), \quad \text{Realizes}(x) \Rightarrow \text{Health}_{\mathfrak{S},r}(x; d, t).$$

¹Formal counterpart: `lean/Tier33/Health/Paper/DefinitionAuthority.lean::core_health_normal_form`.

3 Lawful responses precede viable continuations

Define

$$\mathcal{L}_{d,t}(x) := \{(y, \gamma) \mid \gamma \in \text{Hist}(d, t, x, y), \text{Lawful}(\gamma)\}.$$

The forgetful map

$$\iota : \mathcal{V}_{d,t}(x) \rightarrow \mathcal{L}_{d,t}(x)$$

maps each viable continuation to its endpoint, history, and lawfulness data in $\mathcal{L}_{d,t}(x)$.

Theorem 3.1 (Exact constitutive subfamily).

ι is injective,

$$\boxed{\gamma \in \text{range}(\iota) \iff \text{Continues}(\gamma),} \quad (11)$$

and therefore

$$\boxed{\mathcal{V}_{d,t}(x) \simeq \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\}.} \quad (12)$$

2

The finite model exhibits

$$\exists \gamma \in \mathcal{L}_{d,t}(x), \quad \neg \text{Continues}(\gamma),$$

which establishes strict separation between lawfulness and constitutive continuation.

4 Constitutive response representation and capacity recovery

For a history-outcome predicate P , let

$$\boxed{\mathcal{G}_P(x; d, t) = (C_P^+(x; d, t), C_P^-(x; d, t))} \quad (13)$$

denote selected positive and negative outcome representations of lawful responses. The channels may use different observation and collection maps; neither is assumed to preserve response identity.

For

$$P = \text{Continues},$$

the positive channel uses the same viable histories, positive observer, and collector as (9).

Theorem 4.1 (Conservative-extension certificate).

$$\boxed{\pi_+ \mathcal{G}_{\text{Continues}}(x; d, t) = \text{Capacity}_{\mathfrak{S}}(x; d, t).} \quad (14)$$

3

²Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/LawfulResponseFamily.lean::viable_embedding_injective;viable_image_iff_continues;viableEquivContinuingLawful.`

³Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/ResponseProfile.lean::successAtoms_eq_observedAtoms;positive_projection_eq_capacity.`

The proof is the composition

$$\mathcal{V}_{d,t}(x) \simeq \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\} \implies \text{Obs}_{d,t}^+(x) = \text{Obs}_{d,t}(x) \implies \pi_+ \mathcal{G}_{\text{Continues}} = \text{Capacity}_{\mathfrak{S}}.$$

Hence

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \text{Adequate}_{\mathfrak{S},r}(\pi_+ \mathcal{G}_{\text{Continues}}(x; d, t))}. \quad (15)$$

Equation (14) certifies compatibility of the enriched response representation with the established core.

5 Requirement, scenario, and horizon relativity

The finite fixture of Appendix A fixes one perturbed state P and computes its health under multiple query indices. For the stable-reserve requirement,

	$t = \text{now}$	$t = \text{long}$
$d = \text{mild}$	$\neg \text{Health}_{\mathfrak{S},r_{\text{stable}}}(P; d, t)$	$\text{Health}_{\mathfrak{S},r_{\text{stable}}}(P; d, t)$
$d = \text{severe}$	—	$\neg \text{Health}_{\mathfrak{S},r_{\text{stable}}}(P; d, t)$

(16)

4

At fixed

$$(x, d, t) = (P, \text{mild}, \text{now}),$$

the weaker any-continuation requirement gives

$$\boxed{\text{Health}_{\mathfrak{S},r_{\text{any}}}(P; \text{mild}, \text{now}), \quad \neg \text{Health}_{\mathfrak{S},r_{\text{stable}}}(P; \text{mild}, \text{now})}. \quad (17)$$

5

Thus the generic object remains $\text{Health}_{\mathfrak{S},r}(x; d, t)$, with specification, requirement, scenario, and horizon retained as explicit mathematical indices.

6 Requirement-visible health-capacity quotient

Let $\mathcal{Q}$ be the declared health-requirement language over a domain capacity carrier Cap .

Definition 6.1 (Requirement signature). For $c \in \text{Cap}$,

$$\boxed{\sigma_{\mathcal{Q}}(c) : \mathcal{Q} \rightarrow \text{Prop}, \quad \sigma_{\mathcal{Q}}(c)(r) := \text{Adequate}_r(c)}. \quad (18)$$

Definition 6.2 (Requirement-visible capacity equivalence).

$$\boxed{c_1 \sim_{\mathcal{Q}} c_2 \iff \forall r \in \mathcal{Q}, \quad \text{Adequate}_r(c_1) \leftrightarrow \text{Adequate}_r(c_2)}. \quad (19)$$

⁴Formal counterpart: `lean/Tier33/Health/Paper/DynamicFunctionalityFixture.lean::health_is_horizon_relative;health_is_scenario_relative.`

⁵Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/RequirementSignature.lean::RequirementFixture.requirement_changes_health.`

The relation $\sim_Q$ is an equivalence relation, and

$$c_1 \sim_Q c_2 \iff \sigma_Q(c_1) = \sigma_Q(c_2). \quad (20)$$

6

Define

$$\text{HealthCap}_Q := \text{Cap} / \sim_Q. \quad (21)$$

The term “requirement-visible” is exact: the quotient identifies precisely those raw-capacity distinctions that every declared requirement treats identically. Its interpretation is relative to Q . Enlarging or refining the requirement language can refine the quotient.

Every adequacy predicate descends.

Theorem 6.3 (Adequacy descent). *For each $r \in Q$, there exists*

$$\overline{\text{Adequate}}_r : \text{HealthCap}_Q \rightarrow \text{Prop}$$

such that

$$\overline{\text{Adequate}}_r([c]) \iff \text{Adequate}_r(c). \quad (22)$$

7

The prospective-health family therefore descends with present realization retained as an independent conjunct.

Theorem 6.4 (Health descent).

$$\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \overline{\text{Adequate}}_r([\text{Capacity}_{\mathfrak{S}}(x; d, t)]). \quad (23)$$

8

The signature itself descends to

$$\bar{\sigma}_Q : \text{HealthCap}_Q \rightarrow (Q \rightarrow \text{Prop}),$$

and the induced map is injective. Its range is exactly the attainable range of raw requirement signatures. Each quotient class is therefore characterized by the complete pattern of answers available to the declared health-requirement language.

⁶Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/RequirementSignature.lean::capacityEquivalent_iff_signature_equal`.

⁷Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::adequacy_descends`.

⁸Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::health_descends`.

7 Sufficiency and the universal property

Let

$$E : \text{Cap} \rightarrow Z.$$

Define requirement sufficiency by

$$\boxed{E(c_1) = E(c_2) \implies c_1 \sim_{\mathcal{Q}} c_2.} \quad (24)$$

Theorem 7.1 (Requirement-visible capacity universal property). *Condition (24) is equivalent to the existence of a unique*

$$\bar{E} : \text{range}(E) \rightarrow \text{HealthCap}_{\mathcal{Q}}$$

such that

$$\boxed{\bar{E}(E(c)) = [c]_{\mathcal{Q}}} \quad (25)$$

for every $c \in \text{Cap}$.

9

On the attained range,

$$\boxed{\bar{E}(E(c)) := [c]_{\mathcal{Q}}.}$$

Well-definedness is exactly

$$E(c) = E(c') \xrightarrow{(24)} c \sim_{\mathcal{Q}} c' \implies [c]_{\mathcal{Q}} = [c']_{\mathcal{Q}}.$$

Conversely,

$$E(c_1) = E(c_2) \implies \bar{E}(E(c_1)) = \bar{E}(E(c_2)) \implies [c_1]_{\mathcal{Q}} = [c_2]_{\mathcal{Q}} \implies c_1 \sim_{\mathcal{Q}} c_2.$$

Equivalently, with

$$\begin{aligned} c_1 \equiv_E c_2 &\iff E(c_1) = E(c_2), \\ \boxed{E \text{ requirement-sufficient} &\iff \equiv_E \subseteq \sim_{\mathcal{Q}}.} \end{aligned} \quad (26)$$

Thus the quotient map

$$c \mapsto [c]_{\mathcal{Q}}$$

induces the coarsest requirement-sufficient partition of Cap in the standard refinement order.

8 Strictness of the requirement-visible quotient

The finite dynamic fixture supplies a strict quotient witness using capacities generated by the exact viable-continuation families.

Let

$$c_{\text{quiet}} = \{\text{stable}\}, \quad c_{\text{varied}} = \{\text{stable}, \text{exposed}\}.$$

⁹Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::minimal_capacity_universal_property`.

These capacities are distinct:

$$c_{\text{quiet}} \neq c_{\text{varied}}.$$

For the declared requirement family

$$\mathcal{Q} = \{r_{\text{any}}, r_{\text{stable}}\},$$

where

$$r_{\text{any}}(c) \iff \exists a, a \in c$$

and

$$r_{\text{stable}}(c) \iff \text{stable} \in c,$$

both capacities return the same answer to every declared requirement. Therefore

$$\boxed{[c_{\text{quiet}}] = [c_{\text{varied}}] \text{ in } \text{HealthCap}_{\mathcal{Q}}.} \quad (27)$$

10

The same fixture also contains a genuine requirement-visible distinction. The perturbed current-horizon capacity lacks stable reserve, and the baseline current-horizon capacity contains stable reserve. Their quotient classes are therefore distinct:

$$\boxed{[\text{Capacity}(P; \text{mild}, \text{now})] \neq [\text{Capacity}(B; \text{mild}, \text{now})].} \quad (28)$$

11

Equations (27)–(28) give

$$\boxed{\begin{array}{l} \text{requirement-invisible difference} \mapsto \text{one quotient class,} \\ \text{requirement-visible difference} \mapsto \text{distinct classes.} \end{array}}$$

9 Health-presentation invariance

Fix a requirement r . Let

$$f : X \rightarrow Y, \quad \kappa : \text{Cap}_X \rightarrow \text{Cap}_Y$$

satisfy

$$\text{Realizes}_X(x) \iff \text{Realizes}_Y(fx), \quad (29)$$

$$\kappa(\text{Capacity}_X(x; d, t)) = \text{Capacity}_Y(fx; d, t), \quad (30)$$

$$\text{Adequate}_{X,r}(c) \iff \text{Adequate}_{Y,r}(\kappa c). \quad (31)$$

Proposition 9.1 (Health-presentation invariance).

$$\boxed{\text{Health}_{X,r}(x; d, t) \iff \text{Health}_{Y,r}(fx; d, t).} \quad (32)$$

¹⁰Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::RequirementFixture.strict_generated_viable_capacity_quotient.`

¹¹Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/MinimalCapacityQuotient.lean::RequirementFixture.quotient_retains_requirement_difference.`

12

The hypotheses are essential. The finite coarse-graining control gives

$$\boxed{\text{realization preserved} \not\Rightarrow \text{health preserved.}} \quad (33)$$

13

10 An information boundary: successful capacity and robustness

For the exhibited persistence systems S_0, S_1 ,

$$C_{\text{pers}}^+(S_0) = C_{\text{pers}}^+(S_1),$$

and their robust-persistence statuses are

$$\text{RobustHomeostaticPersistence}(S_0), \quad \neg \text{RobustHomeostaticPersistence}(S_1).$$

Theorem 10.1 (Successful persistence information is insufficient in general for robustness).

$$\boxed{\neg \exists f : C_{\text{pers}}^+ \rightarrow \mathbf{Prop}, \forall i \in \{0, 1\}, \quad f(C_{\text{pers}}^+(S_i)) \iff \text{RobustHomeostaticPersistence}(S_i).} \quad (34)$$

14

Equality of the positive carriers forces every f on that carrier to agree on S_0, S_1 . The robust-persistence predicate separates the two systems. Therefore

$$\boxed{\text{positive prospective representation} \not\supseteq \text{all failure-sensitive response information.}}$$

The theorem establishes generic insufficiency on the exhibited family; restricted classes may admit a factorization.

11 Epistemic and empirical boundary

For the finite mild/current query, let

$$O_{\text{present}} : X \rightarrow \{\text{realizing}, \text{nonrealizing}\}.$$

The fixture contains B, P satisfying

$$O_{\text{present}}(B) = O_{\text{present}}(P),$$

with health values

$$\text{Health}(B; \text{mild}, \text{now}), \quad \neg \text{Health}(P; \text{mild}, \text{now}).$$

¹²Formal counterpart:

`lean/Tier33/Health/Paper/IndexedInvariance.lean::health_iff_of_healthRelevantPresentationMap.`

¹³Formal counterpart:

`lean/Tier33/Health/Paper/SeparationControls.lean::realization_preserving_map_can_destroy_health.`

¹⁴Formal counterpart: `lean/Tier33/Health/Paper/ResponseSemantics/RobustFactorization.lean::`

`ProjectionObstruction.robust_persistence_does_not_factor_through_viable_capacity.`

Theorem 11.1 (Present-realization observation insufficiency).

$$\boxed{\neg \text{HealthSufficient}(O_{\text{present}}, \text{Health}(\cdot; \text{mild}, \text{now}))}. \quad (35)$$

15

For

$$O_{\text{Health}}(x) = (O_{\text{present}}(x), \text{Capacity}_{\mathfrak{S}}(x; \text{mild}, \text{now})),$$

$$\boxed{\text{HealthSufficient}(O_{\text{Health}}, \text{Health}(\cdot; \text{mild}, \text{now}))}. \quad (36)$$

16

Hence

$$\boxed{\text{formal health} \neq \text{recoverability of health from a chosen measurement.}} \quad (37)$$

Physiological sufficiency, diagnostic validity, and empirical identification require additional evidence.

12 Related work and novelty positioning

The framework intersects several established traditions. The contribution is best understood by locating those ingredients precisely and then stating what is assembled here.

12.1 Dynamic and action-relative conceptions of health

Dynamic accounts of health already emphasize adaptation, self-management, and the ability to pursue or restore valued functioning under changing conditions. Huber et al. emphasize adaptation and self-management [1]; Nordenfelt develops an action-theoretic relation between health and the ability to realize vital goals [2]. Boorse’s biostatistical account supplies a complementary tradition centered on statistically normal biological function [3]. The novelty claim here concerns a typed prospective semantics in which present constitution, continuation, capacity, and adequacy are separate mathematical interfaces.

12.2 Viability and invariant-set theory

Viability theory studies evolutions that remain compatible with constraints, together with viability kernels, capture, regulation, and restoration [4]. Classical viability kernels classify initial states. The object $\mathcal{V}_{d,t}(x)$ used here is a rooted proof-carrying family of viable histories. Under a pointwise constraint interpretation of Continues, nonemptiness of this family is viability-like; retaining the individual histories allows a domain to derive a richer capacity representation.

Invariant-set and robust-control theory distinguish existence of successful control from stronger disturbance-robust requirements [5]. Section 10 uses this quantifier distinction to establish the

¹⁵Formal counterpart: `lean/Tier33/Health/Paper/DynamicFunctionalityFixture.lean::present_observation_not_health_sufficient.`

¹⁶Formal counterpart: `lean/Tier33/Health/Paper/DynamicFunctionalityFixture.lean::healthPresentation_sufficient.`

paper’s robustness boundary. Robust, controlled, cost-sensitive, and risk-sensitive capacity form downstream extensions of the present foundation.

12.3 Predicate abstraction, observational equivalence, and sufficiency

The requirement signature

$$\sigma_Q(c)(r) = \text{Adequate}_r(c)$$

is closely related to predicate abstraction and observational equivalence [6, 7, 8]. In the present framework, this established quotient principle is instantiated by the formal health-requirement language.

The universal property of HealthCap_Q also has the familiar shape of minimal sufficiency: any representation sufficient for the selected family of questions must determine the coarser task-relevant class [9]. Recent work on Bayes-sufficient representations provides a closely related decision-relative quotient viewpoint [10, 11]. The present theorem is deterministic and proposition-valued, with the distinguishing language supplied by prospective-health requirements.

12.4 Machine-checked systems theory

Machine-checked reasoning about dynamical and control systems is an active area, including recent Lean formalizations of stability and small-gain reasoning [12]. The contribution here is the formalization of this particular health architecture together with its explicit authority boundaries.

12.5 Contribution statement

The contribution is the machine-checked integration

$$\boxed{\begin{array}{l} \mathcal{L} \xrightarrow{\text{constitutive restriction}} \{\gamma \in \mathcal{L} \mid \text{Continues}(\gamma)\} \simeq \mathcal{V}, \\ \mathcal{V} \rightarrow \text{Obs} \rightarrow \text{Capacity} \rightarrow \text{HealthCap}_Q \rightarrow \text{Health}_{\mathfrak{S},r}. \end{array}}$$

together with the exact lawful/viable identification, the requirement-visible universal property, strict relativity witnesses, qualified presentation invariance, and two explicit boundary theorems. The novelty claim is restricted to this integrated architecture and theorem package; the constituent viability, quotient, observational-equivalence, and minimal-sufficiency principles retain their established provenance.

13 Discussion

The finite model and quotient construction establish

$$\begin{array}{c} \text{Health}_{\mathfrak{S},r}(x; d, t) \text{ is genuinely indexed by } (r, d, t), \\ [c_1]_{\mathcal{Q}} = [c_2]_{\mathcal{Q}} \iff \forall r \in \mathcal{Q}, \text{Adequate}_r(c_1) \leftrightarrow \text{Adequate}_r(c_2). \end{array}$$

The two boundary theorems give

$$\begin{array}{c} \text{Capacity} \not\approx \text{failure-sensitive robustness}, \\ \text{present-realization observation} \not\approx \text{measurement sufficiency for Health}. \end{array}$$

These are the precise stopping conditions of the foundational paper. Downstream work extends query governance, response structure, and empirical identification and retains the prospective-health normal form in Equation (1).

13.1 A measurand for health

The formal health predicate also fixes the logical order of a measurement problem. For a specified query,

$$\text{Health}_{\mathfrak{S},r}(x; d, t)$$

is the system-level proposition to be determined. A candidate observation

$$O : X \rightarrow Y$$

is health-sufficient for that query only if its fibers are homogeneous with respect to the health predicate:

$$O(x_1) = O(x_2) \implies (\text{Health}_{\mathfrak{S},r}(x_1; d, t) \leftrightarrow \text{Health}_{\mathfrak{S},r}(x_2; d, t)).$$

The finite observation control in Section 11 shows that present-realization observation can fail this condition, while the corresponding presentation containing present realization and exact prospective capacity is sufficient for the fixed query.

The requirement-visible quotient clarifies the information requirement. For a fixed formal requirement language $\mathcal{Q}$,

$$c_1 \sim_{\mathcal{Q}} c_2 \iff \forall r \in \mathcal{Q}, \text{Adequate}_r(c_1) \leftrightarrow \text{Adequate}_r(c_2),$$

so a representation sufficient for the complete requirement language must determine

$$\text{HealthCap}_{\mathcal{Q}}.$$

A health measurement therefore need not reconstruct the complete physiological state or the complete viable-response family. It must preserve the distinctions required to determine the specified health propositions.

This ordering separates the measurand from its observation:

$$\text{health proposition} \longrightarrow \text{required health-relevant information} \longrightarrow \text{candidate observation}.$$

The foundational paper establishes the first two objects and one finite sufficiency boundary. Measurement identification, calibration, uncertainty, and empirical validation form the downstream measurement theory.

14 Scope boundaries

The theorem prover works relative to the declared formal semantics:

declared formal semantics $\implies$ machine-checked consequences about $\mathbf{Health}_{\mathfrak{S},r}(x; d, t)$.

Biological and clinical correctness remain domain and empirical obligations.

The following remain downstream scientific or formal tasks:

scientific licensing of requirement/scenario/horizon families,
robustness, probability, risk, cost, and control semantics,
physiological measurement sufficiency,
diagnostic and treatment authority,
empirical comparison with competing health assessments.

The robustness theorem establishes nonfactorization on the exhibited family. The observation theorem is scoped to one finite query. Physiological measurement models require a separate validation layer.

15 Conclusion

The foundational architecture is

$$\boxed{\begin{array}{l} \mathcal{L}_{d,t}(x) \xrightarrow{\text{constitutive restriction}} \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\} \simeq \mathcal{V}_{d,t}(x), \\ \mathcal{V}_{d,t}(x) \xrightarrow{o} \text{Obs}_{d,t}(x) \xrightarrow{\text{collect}} \text{Capacity}_{\mathfrak{S}}(x; d, t) \xrightarrow{[-]_{\mathcal{Q}}} \text{HealthCap}_{\mathcal{Q}}. \end{array}} \quad (38)$$

with

$$\boxed{\begin{array}{l} \iota : \mathcal{V}_{d,t}(x) \hookrightarrow \mathcal{L}_{d,t}(x), \\ \text{im } \iota = \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\}, \quad \mathcal{V}_{d,t}(x) \simeq \{\gamma \in \mathcal{L}_{d,t}(x) \mid \text{Continues}(\gamma)\}. \end{array}} \quad (39)$$

and

$$\boxed{\text{Health}_{\mathfrak{S},r}(x; d, t) \iff \text{Realizes}_{\mathfrak{S}}(x) \wedge \overline{\text{Adequate}_r}([\text{Capacity}_{\mathfrak{S}}(x; d, t)]).} \quad (40)$$

The completeness claim belongs to $\mathcal{V}_{d,t}(x)$. The capacity is the specification's downstream representation after observation and collection.

For a fixed formal requirement language,

$$\boxed{c_1 \sim_{\mathcal{Q}} c_2 \iff \forall r \in \mathcal{Q}, \text{Adequate}_r(c_1) \leftrightarrow \text{Adequate}_r(c_2), \quad \text{HealthCap}_{\mathcal{Q}} = \text{Cap} / \sim_{\mathcal{Q}}.} \quad (41)$$

The finite witnesses establish requirement, scenario, and horizon relativity, strict quotient collapse, and retained requirement-visible distinctions. Presentation invariance holds under explicit transport of realization, capacity, and adequacy.

Finally,

$$\boxed{\begin{array}{l} \text{positive prospective representation} \not\Rightarrow \text{robust persistence in general,} \\ \text{present-realization observation} \not\Rightarrow \text{sufficient observation of prospective health.} \end{array}}$$

These boundaries define the stopping point of Paper I.

A Finite model details

The finite functionality fixture uses states

$$B = \text{baseline}, \quad P = \text{perturbed}, \quad R = \text{recovered}, \quad A = \text{adapted}, \quad F = \text{failed}.$$

Realization holds exactly on B, P, R, A .

The history family contains self histories for all states; a mild baseline-to-perturbed history; mild/long recovery and adaptation histories from P ; and a severe/long history from P to F .

The severe failure history is lawful and fails constitutive continuation.

Observed capacity atoms are stable and exposed.

The exact capacities used in the manuscript are:

$$\text{Capacity}(B; \text{mild}, \text{now}) = \{\text{stable}, \text{exposed}\},$$

$$\text{Capacity}(P; \text{mild}, \text{now}) = \{\text{exposed}\},$$

$$\text{Capacity}(P; \text{mild}, \text{long}) = \{\text{exposed}, \text{stable}\},$$

$$\text{Capacity}(P; \text{severe}, \text{long}) = \{\text{exposed}\},$$

and

$$\text{Capacity}(R; \text{mild}, \text{now}) = \{\text{stable}\}.$$

The formal requirement language used for quotient strictness contains:

$$r_{\text{any}}(c) \iff \exists a, a \in c,$$

and

$$r_{\text{stable}}(c) \iff \text{stable} \in c.$$

Both

$$\{\text{stable}\}$$

and

$$\{\text{stable}, \text{exposed}\}$$

satisfy the same two requirement queries and therefore define the same class in $\text{HealthCap}_{\mathcal{Q}}$.

B Reader theorem map

ID	Section	Mathematical statement	Formal counterpart
CCH-01	Health core	$\text{Health}_{\mathfrak{S},r} = \text{Realizes}_{\mathfrak{S}} \wedge \text{Adequate}_{\mathfrak{S},r}(\text{Capacity}_{\mathfrak{S}})$.	<code>DefinitionAuthority.core_health_normal_form</code>
CCH-02	Lawful response	Viable continuations inject into lawful responses.	<code>viable_embedding_injective</code>
CCH-03	Lawful response	The viable image is exactly the continuing lawful subtype.	<code>viable_image_iff_continues</code>
CCH-04	Lawful response	Exact equivalence with the continuing lawful subtype.	<code>viableEquivContinuingLawful</code>
CCH-05	Capacity recovery	Positive constitutive outcome atoms equal the established viable observed atoms.	<code>successAtoms_eq_observedAtoms</code>
CCH-06	Capacity recovery	Positive constitutive projection recovers domain capacity.	<code>positive_projection_eq_capacity</code>
CCH-07	Requirement relativity	One state has different health judgments under two formal requirements.	<code>RequirementFixture.requirement_changes_health</code>
CCH-08	Dynamic fixture	Health is scenario-relative in the complete finite model.	<code>DynamicFunctionalityFixture.health_is_scenario_relative</code>
CCH-09	Dynamic fixture	Health is horizon-relative in the complete finite model.	<code>DynamicFunctionalityFixture.health_is_horizon_relative</code>
CCH-10	Requirement language	Capacity equivalence is equality of requirement signatures.	<code>capacityEquivalent_iff_signature_equal</code>
CCH-11	Requirement quotient	Every adequacy predicate descends to $\text{HealthCap}_{\mathcal{Q}}$.	<code>adequacy_descends</code>
CCH-12	Requirement quotient	The requirement-relative health family descends to the quotient.	<code>health_descends</code>
CCH-13	Universal property	Every requirement-sufficient encoding uniquely determines the quotient on its attained range.	<code>minimal_capacity_universal_property</code>
CCH-14	Strictness	Distinct generated viable capacities occupy one requirement-visible quotient class.	<code>RequirementFixture.strict_generated_viable_capacity_quotient</code>
CCH-15	Strictness	A formal stable-reserve difference remains separated by the quotient.	<code>RequirementFixture.quotient_retains_requirement_difference</code>
CCH-16	Presentation	Health equivalence follows from a health-relevant presentation map.	<code>IndexedInvariance.health_iff_of_healthRelevantPresentationMap</code>
CCH-17	Presentation control	Realization-preserving coarse-graining can change health.	<code>SeparationControls.realization_preserving_map_can_destroy_health</code>
CCH-18	Robustness boundary	Equal positive persistence information can coexist with opposite robust-persistence status.	<code>ProjectionObstruction.robust_persistence_does_not_factor_through_viable_capacity</code>
CCH-19	Epistemic boundary	Present-only observation has a mixed health fiber for the finite current-horizon query.	<code>DynamicFunctionalityFixture.present_observation_not_health_sufficient</code>
CCH-20	Epistemic boundary	Present-realization observation plus exact prospective capacity is sufficient for the fixed finite query.	<code>DynamicFunctionalityFixture.healthPresentation_sufficient</code>

C Formal authority and verification

The paper-facing response-semantics authority is rooted in

```
lean/Tier33/Health/Paper/ResponseSemantics/Completion.lean,
```

with reproducible axiom inspection in

```
lean/Tier33/Health/Paper/ResponseSemantics/AxiomAudit.lean.
```

The verified paper source is pinned at commit

```
9dd24a73fe0d5dc0f3636d702e259a3aa9a021d2.
```

The checked environment is

<code>Lean 4.28.0-rc1,</code>	<code>mathlib 5352afccd6866369be9de43f5b7ec47203555f44.</code>
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The formal development contains a larger verified response-semantics surface than the theorem basis emphasized in this foundational manuscript. The principal paper-facing results used here are the lawful/viable embedding and subtype equivalence, constitutive positive-projection recovery, requirement signature and quotient descent, universal property, strict quotient witnesses, finite scenario/horizon/requirement countermodels, presentation invariance, the robust-persistence obstruction, and the finite observation-sufficiency controls.

The committed response-semantics axiom audit reports the kernel dependency set

<code>{propext, Quot.sound, Classical.choice}</code>
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across the audited theorem surface. The source audit confirms kernel-checked proofs throughout the new response-semantics modules and excludes proof `sorry`, proof `admit`, custom project axioms, `native_decide`, unconstrained completion Booleans, and caller-selected favorable response banks.

Classical choice is used in the inherited quotient-range representative construction. Requirement sufficiency proves that the resulting quotient class is independent of the chosen representative.

Scope. Independent reproduction requires the pinned source tree, its `lean-toolchain`, Lake configuration and manifest, and the paper-facing build commands. The checked environment uses a Lean release candidate; durable archival reproduction can preserve the complete pinned environment or record a successful replay under the corresponding stable release.

D Theorem-classification guide

Class	Examples
Definition / interface	Prospective-health normal form and typed interfaces.
Representation theorem	Lawful/viable subtype equivalence; conservative-extension certificate; presentation transport.
Countermodel / obstruction	Lawful noncontinuation; requirement/scenario/horizon relativity; health-destroying coarse presentation; robust nonfactorization; present-observation insufficiency.
Standard principle applied here	Requirement-signature equivalence, predicate-induced quotienting, and sufficiency/minimality factorization.
Universal-property theorem	Requirement-visible health-capacity quotient and unique determination by every requirement-sufficient encoding on its attained range.
Empirical boundary	Physiological interpretation, measurement sufficiency, diagnostic authority, and predictive superiority remain external obligations.

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